

ESP32 client

Python server

Generate ephemeral X25519
and ML-KEM-768 keys
HMAC the unsigned request

POST /handshake: 1,249-byte ClientHello

Verify request HMAC
X25519 + ML-KEM encapsulation
HKDF; generate SID
HMAC request + unsigned response

HTTP response: 1,168-byte ServerHello

Verify transcript HMAC
X25519; reject all-zero secret
ML-KEM decapsulation; HKDF

POST /api/telemetry: SID + IV + ciphertext + tag

Check counter; authenticate/decrypt
Update counter after valid processing
Encrypt acknowledgment

Encrypted acknowledgment

One application key-exchange request/response; TCP setup is separate.
No Finished flight; telemetry shares one key across directions and uses no AAD.
Rotate after 50 successful responses; three consecutive errors also end the session.